Problem Set 5 (due 11:59 pm, 17 November 2025)

You are allowed to consult AI, but direct copying codes from AI is prohibited.

The solution should be submitted as a Jupyter Notebook. Use "Heading" to label the question parts, "Code" to write the code, and "Markdown" to summarize the results and answer the questions in text.

Online Learning in Soft Committee Machines

This problem should be solved with Python. Since random numbers are used to generate the teacher and the learning examples, you should assign the seed number so that you (and the IA) can check your work, and you should measure results averaged over at least 30 samples.

(1) Construct a teacher network that is a soft committee machine with $N = 1{,}000$ inputs and 3 hidden units. Generate the weights B_{ni} feeding hidden unit n from input i using Gaussian distributions (normal distributions) with mean 0 and variance $1/N$.

The weights from the hidden units to the output are set to 1.

For input vector $\boldsymbol{\xi}$, the field y_n at hidden unit n is given by

$$y_n = \sum_{i=1}^{N} B_{ni}\xi_i.$$

The activation at hidden unit n is given by

$$g(y_n) = \operatorname{erf}\left(\frac{y_n}{\sqrt{2}}\right).$$

In Python, you may use the function `scipy.special.erf()` if the argument of the function is an array.

The output of the soft committee machine is given by

$$\sigma = \sum_{n=1}^{3} g(y_n).$$

(2) Compute the overlaps T_{nm} between the weights feeding hidden units n and m defined as

$$T_{nm} = \sum_{i} B_{ni}\, B_{mi},$$

and check whether they are closed to δ_{nm}.

(2) Construct a student network with the same architecture as the teacher network. The student weights feeding hidden unit n from input i are denoted as J_{ni} and are initialized using Gaussian distributions with mean 0 and variance $1/N^2$. For input vector $\boldsymbol{\xi}$, the student field of hidden unit n is denoted as

$$x_n = \sum_{i=1}^{N} J_{ni} \xi_i.$$

The weights from the hidden units to the output are set to 1.

(3) The example at each step consists of an input vector $\boldsymbol{\xi}$, and the output σ prescribed by the teacher network. Each component ξ_i of the input vector is an i.i.d. variable generated from the Gaussian distribution with mean 0 and variance 1. The learning rate is η/N and the learning time α advances by $1/N$ at each step. Implement the online learning algorithm for the student using

$$\Delta J_{nj} = \frac{\eta}{N} g'(x_n) \left[\sum_{m=1}^{3} g(y_m) - \sum_{m=1}^{3} g(x_m) \right] \xi_j.$$

Note that

$$g'(x) = \frac{\sqrt{2}}{\sqrt{\pi}} e^{-x^2/2}.$$

You should select a low learning rate η/N in the specialization regime. Perform the simulation up to $\alpha = 300$.

(4) At each instant of time separated by $\Delta\alpha = 1$, generate the overlaps given by

$$Q_{nm} = \sum_{i=1}^{N} J_{ni} J_{mi}, \qquad R_{nm} = \sum_{i=1}^{N} J_{ni} B_{mi}.$$

(5) Plot the parameters R_{nm} as a function of α on the same graph. For the convenience of the reader, please include the legend referring to the 9 curves. Please also specify which student hidden unit is specialized to represent which teacher hidden unit.

(6) At each instant of time separated by $\Delta\alpha = 1$, generate an independent set of 1,000 examples and compute the generalization error given by the average of

$$\varepsilon = \frac{1}{2} \left[\sum_{m=1}^{3} g(y_m) - \sum_{m=1}^{3} g(x_m) \right]^2.$$

(7) Plot the generalization error as a function of α. For comparison, plot the theoretical curve on the same graph given by

$$\epsilon_g(\mathbf{J}) = \frac{1}{\pi} \left\{ \sum_{i,k} \arcsin \frac{Q_{ik}}{\sqrt{1+Q_{ii}}\sqrt{1+Q_{kk}}} + \sum_{n,m} \arcsin \frac{T_{nm}}{\sqrt{1+T_{nn}}\sqrt{1+T_{mm}}} \right.$$

$$\left. -2 \sum_{i,n} \arcsin \frac{R_{in}}{\sqrt{1+Q_{ii}}\sqrt{1+T_{nn}}} \right\}.$$